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SUB: Deep learning

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3	14/08	Study of the classifiers with respect to statistical parameters		
4	14/08	Build a simple feed forward neural network to register uc		
5	22/08	Study of activation function and its role		
6	09/09	Gradient descent & backpropagation in deep neural networks		
7	16/09	Build a cnn model classify cat and dog image		
8	9/10	experiment of LSTM		
9	09/10	Build a Recurrent Neural Network		
10	17/10	Perform compression on mnist dataset using Autoencoder		
11	17/10	Experiments using variational Autoencoder (VAE)		
12	26/10	Implement a Deep convolutional GAN (DCGAN) to color image		
		Understand the Architecture of the trained model		

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10. Perform compression on mnist dataset using auto encoder

Aim

To implement an Autoencoder for compressing and reconstructing images from the MNIST dataset, and to study its ability to learn low-dimensional representations.

Objectives

1. To understand the concept the working of autoencoders
2. To perform image compression using the encoder decoder architecture
3. To visualize and analyse reconstructed images and their quality
4. To explore dimensionally reduction through neural networks

Pseudocode

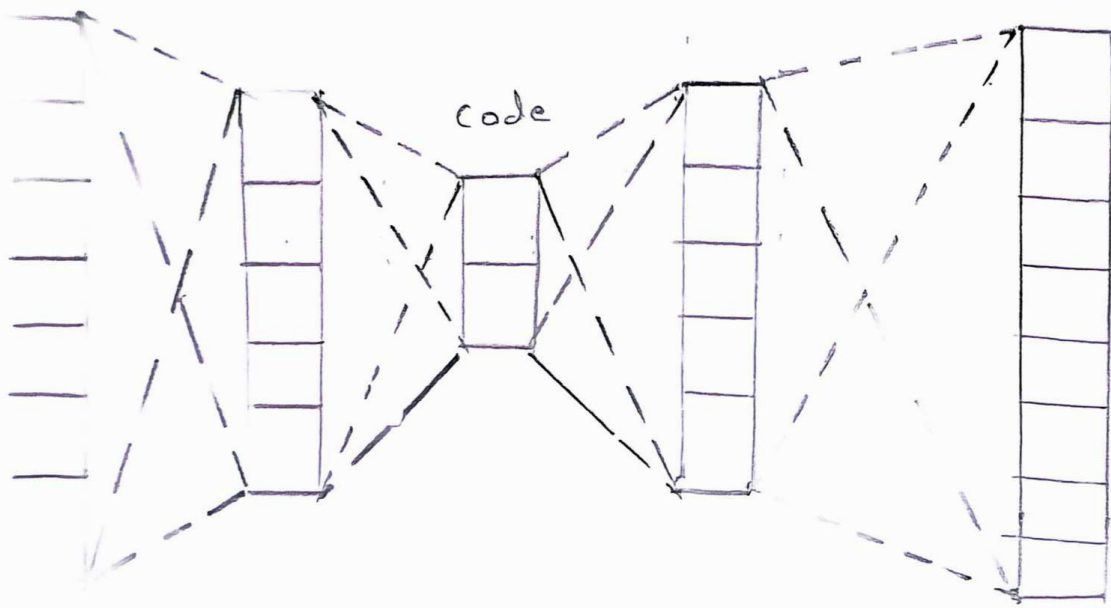
- ~~Start~~
1. Import necessary libraries (Tensorflow/tensor, keras, numpy, matplotlib).
 2. Load MNIST dataset
 - Normalize pixel value (0-1)
 - Flatten images into 784-dimensional vectors
 3. Define Autoencoder architecture
Encoder

Autoencoder

Architecture

Input

Output



Encoder

~~Decoder~~

- Input Layer (784)
 - Dense (128, activation = 'relu',
 - Dense (64, activation = 'relu',
 - Dense (32, activation = 'relu')
- Compressed representation

Decoder
Output Layer (

- Dense (64, activation = 'relu',
- Dense (128, activation = 'relu',
- Dense (256, activation = 'sigmoid')

End

Observation

1. Training Behaviour

- * The autoencoder successfully learned to reconstruct digit images after several epochs
- * Lower learning rates produced smoother convergence

2. Compression Effect:

- * Minor loss of sharpness on fine features was observed due to dimensionality reduction.

3. Visualization:

- * A plot showed strong similarity between original and reconstructed digits

Result

Successfully implemented an autoencoder for compression of MNIST.

Output:

0	0.9519	0.9490	0.7504	480
1	0.9426	0.9692	0.9557	1135
2	0.8175	0.6926	0.7534	1032
3	0.6288	0.5871	0.6074	1010
4	0.6875	0.4817	0.3661	982
5	0.6042	0.5132	0.5552	892
6	0.8322	0.8756	0.8627	958
7	0.8750	0.8716	0.8783	1028
8	0.6165	0.7608	0.6811	974
9	0.5572	0.7334	0.6833	1009
a cen avg				
Macro avg	0.7513	0.7460	0.7439	1000
Weighted avg	0.7555	0.7510	0.7485	1000